

Microeconomics II – Class 11

Slutsky equation



Class summary

- * Slutsky equation
 - * Law of demand
 - * Substitution and income effects for different types of preferences
 - * Distinction between Slutsky's and Hicks' income effects
 - * Demand functions: standard; Slutsky; Hicks or compensated
 - * Slutsky equation with derivatives
- * Readings: Varian (chap. 8 e 4.3)



Law of demand

- * Law of demand:

If the demand for a good increases when income increases, then the demand for that good must decrease when its price increases.

- * Proof using:

- * Slutsky identity $\Delta x_1 \equiv \Delta x_1^s + \Delta x_1^n$

- * Weak Axiom of Revealed Preferences



Proof of the law of demand using Slutsky's substitution effect

- * Proof using Slutsky's identity (and WARP):

- * Using WARP we have proved that $\Delta p_1 > 0 \Rightarrow \Delta x_1^s < 0$

- * Because $\Delta p_1 > 0 \Rightarrow \Delta^- \text{ purchasing power} \Rightarrow \Delta x_1^n < 0$

- * Then, for normal goods $\Delta p_1 > 0 \Rightarrow \begin{cases} \Delta x_1^s < 0 \\ \Delta x_1^n < 0 \end{cases} \Rightarrow \Delta x_1 < 0$

- * Because $\Delta x_1 \equiv \Delta x_1^s + \Delta x_1^n$ then $\Delta p_1 > 0 \Rightarrow \Delta x_1 < 0$



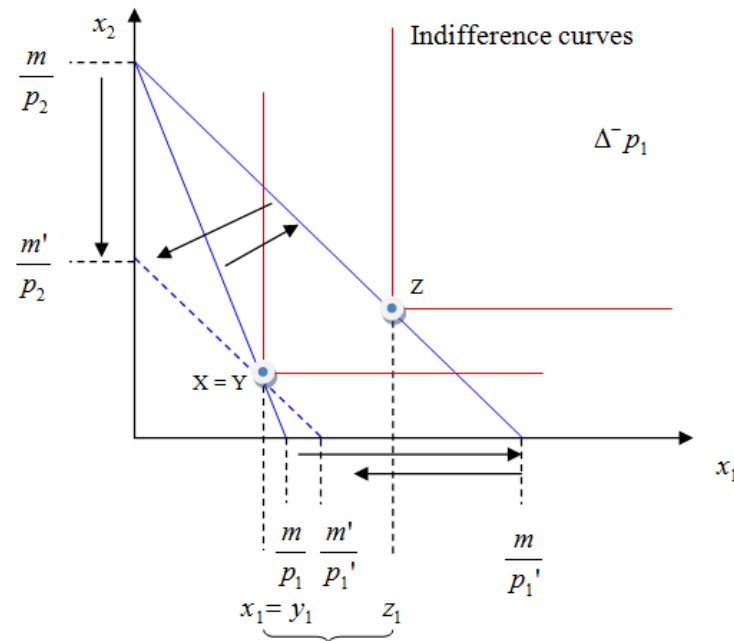
Substitution and income effects for different types of preferences

- * Perfect

complements:

$$u(x_1, x_2) = \min\{ax_1, bx_2\}$$

- * The substitution effect is zero.
- * The entire change in demand is due to the income effect.



$$\Delta x_1 = \Delta x_1^n$$

$$\Delta x_1^s = 0$$



Substitution and income effects for different types of preferences

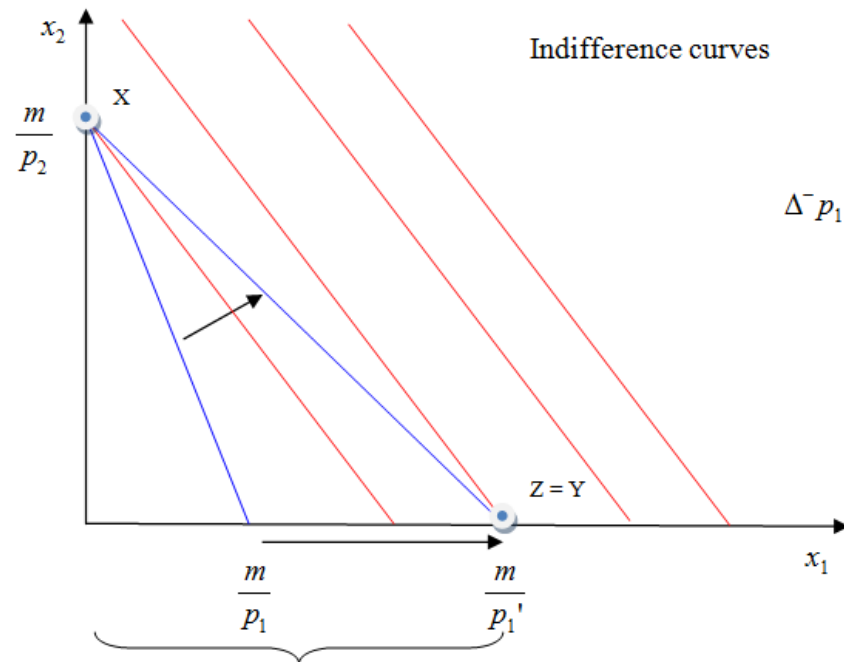
- * Perfect substitutes:

$$u(x_1, x_2) = ax_1 + bx_2$$

- * Case 1:

- * The income effect is zero.

- * The entire change in demand is due to the substitution effect.



$$\Delta x_1 = \Delta x_1^s$$

$$\Delta x_1^n = 0$$



Substitution and income effects for different types of preferences

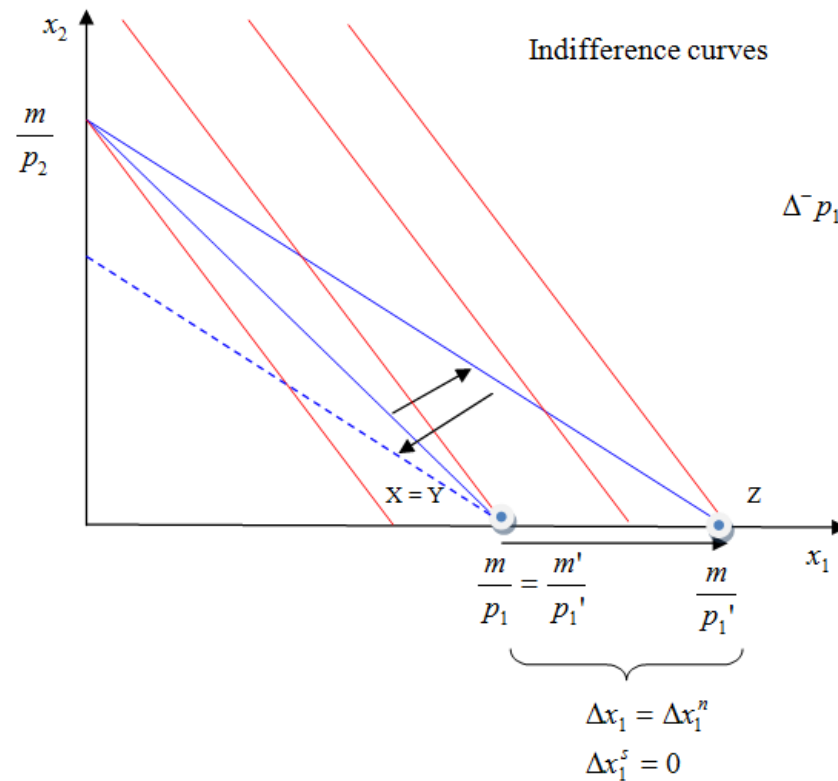
- * Perfect substitutes:

$$u(x_1, x_2) = ax_1 + bx_2$$

- * Case 2:

- * The substitution effect is zero.

- * The entire change in demand is due to the income effect.



Substitution and income effects for different types of preferences

- * Quasilinear preferences:

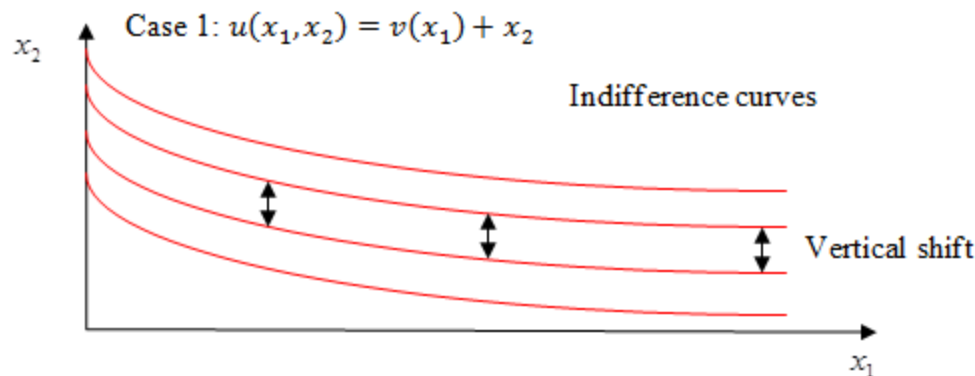
$$u(x_1, x_2) = v(x_1) + x_2$$

Linear part

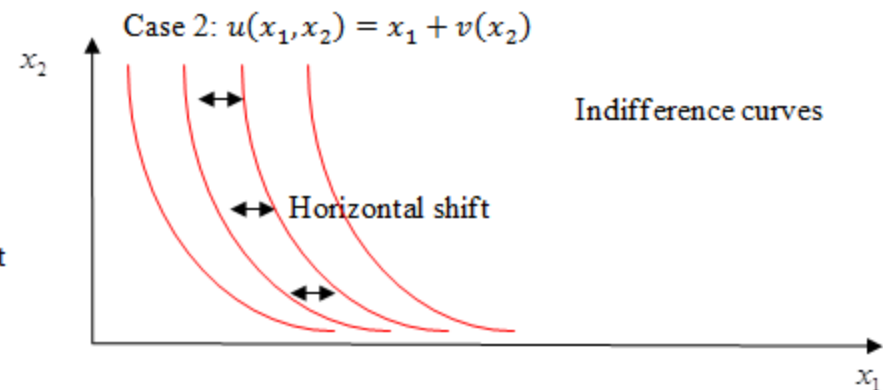
Nonlinear part

- * Indifference curves: $u(x_1, x_2) = k$

$$u(x_1, x_2) = v(x_1) + x_2 \Leftrightarrow x_2 = u(x_1, x_2) - v(x_1) \Leftrightarrow x_2 = k - v(x_1)$$



Along a vertical line all indifference curves have the same slope



Along an horizontal line all indifference curves have the same slope



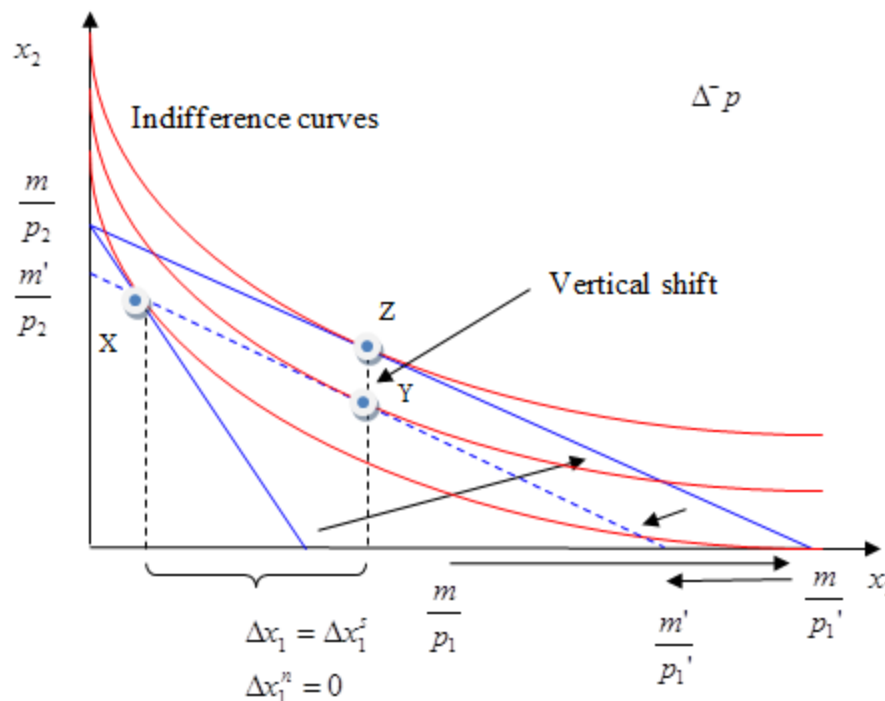
Substitution and income effects for different types of preferences

- * Quasilinear preferences (1):

$$u(x_1, x_2) = v(x_1) + x_2$$

- * The income effect is zero.
- * The entire change in demand is due to the substitution effect.

Case 1: $u(x_1, x_2) = v(x_1) + x_2$ [nonlinearity in the good whose price is changed]



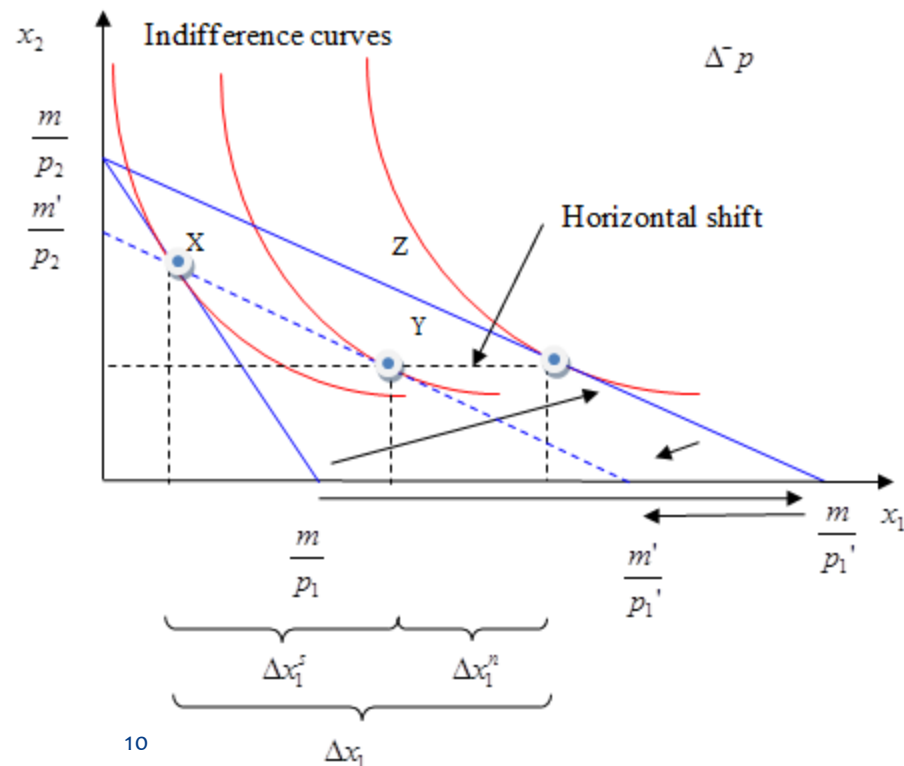
Substitution and income effects for different types of preferences

- * Quasilinear preferences (2):

$$u(x_1, x_2) = x_1 + v(x_2)$$

- * Neither the substitution nor the income effect are null.

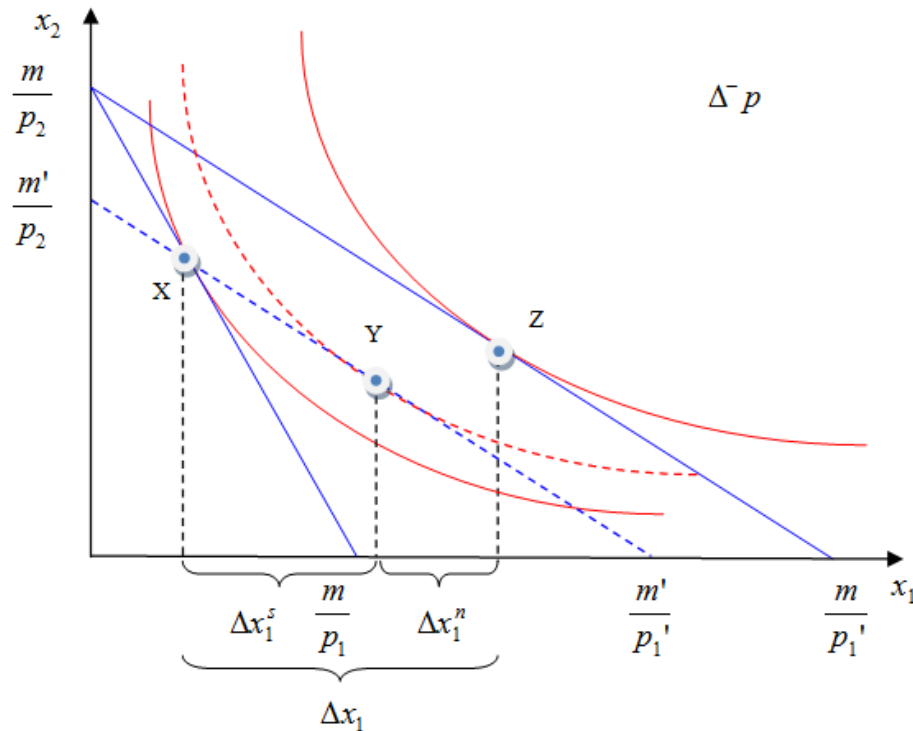
Case 2: $u(x_1, x_2) = x_1 + v(x_2)$



Distinction between Slutsky and Hicks substitution effects

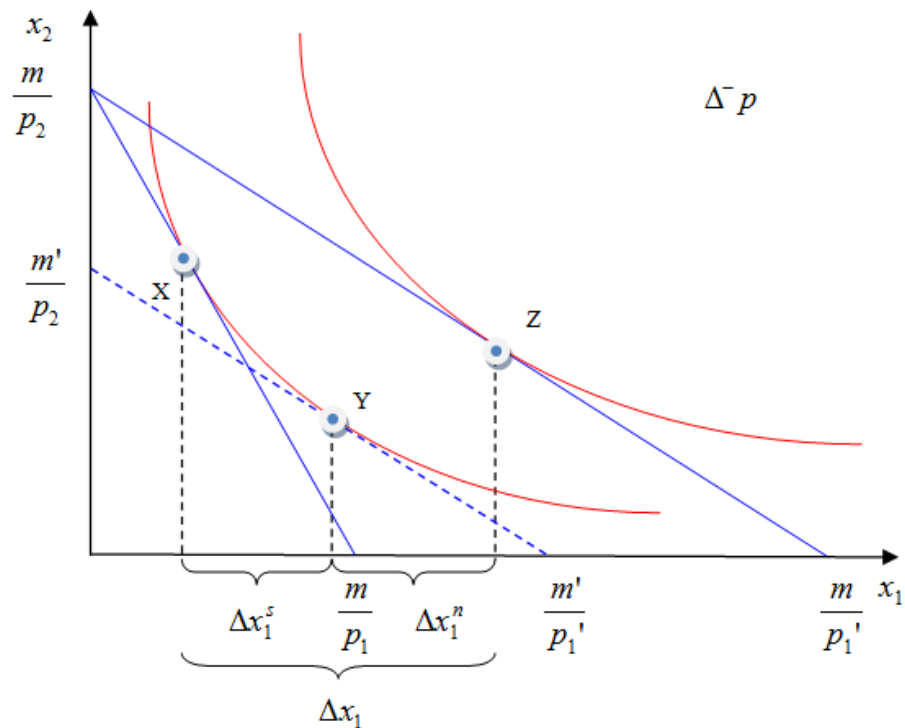
- * Slutsky substitution effect:

- * the purchasing power is kept constant.
- * Income will have to be such that the previously chosen consumption bundle is on the budget constraint (is exactly affordable)



Distinction between Slutsky and Hicks substitution effects

- * Hicks substitution effect:
 - * the utility level is kept constant.
 - * Income will have to be such that the chosen consumption bundle is on the same indifference curve as the previous one.



Hicks Substitution Effect is also negative

Observation	Quantity	Price
X	(x_1, x_2)	(p_1, p_2)
Y	(y_1, y_2)	(q_1, q_2)

* $X \sim Y$ by definition of Hicks substitution effect

$$X \sim Y \Leftrightarrow \neg(X \succ Y) \wedge \neg(Y \succ X) \Leftrightarrow \begin{cases} \neg(p_1 x_1 + p_2 x_2 > p_1 y_1 + p_2 y_2) \\ \neg(q_1 y_1 + q_2 y_2 > q_1 x_1 + q_2 x_2) \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} p_1 x_1 + p_2 x_2 \leq p_1 y_1 + p_2 y_2 \\ q_1 y_1 + q_2 y_2 \leq q_1 x_1 + q_2 x_2 \end{cases} \Rightarrow$$

$$\Rightarrow p_1 x_1 + p_2 x_2 + q_1 y_1 + q_2 y_2 \leq p_1 y_1 + p_2 y_2 + q_1 x_1 + q_2 x_2 \Leftrightarrow$$

$$\Leftrightarrow (q_1 - p_1) \times (y_1 - x_1) + (q_2 - p_2) \times (y_2 - x_2) \leq 0 \Leftrightarrow$$

$$\Leftrightarrow (q_1 - p_1) \times (y_1 - x_1) \leq 0 \Leftrightarrow \Delta p_1 \times \Delta x_1 \leq 0$$

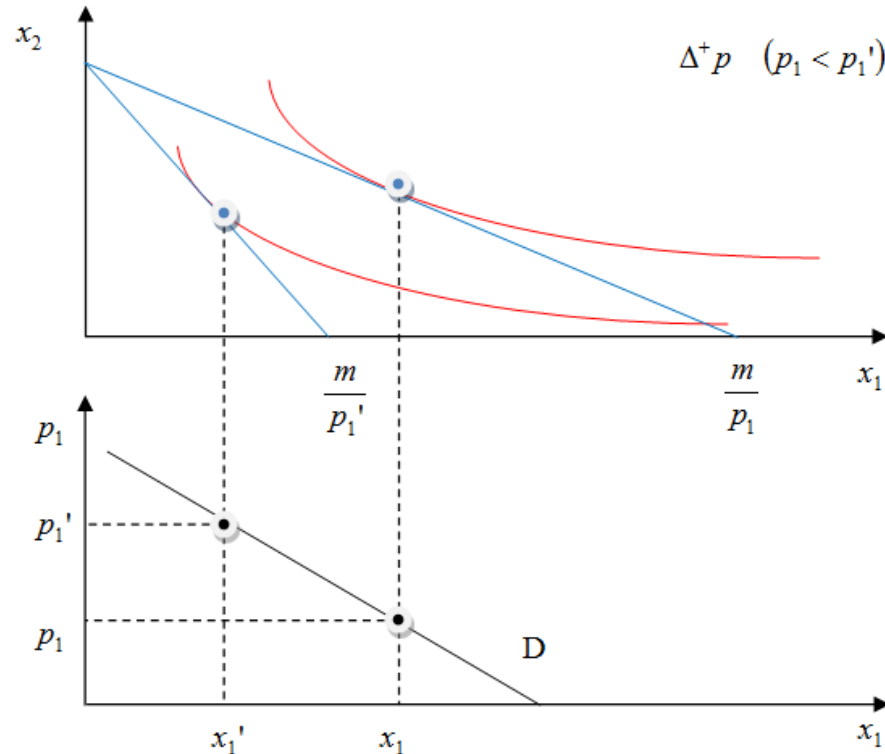
Note: $\neg$ is the negation symbol

* Note that $\Delta p_2 = 0$ by definition



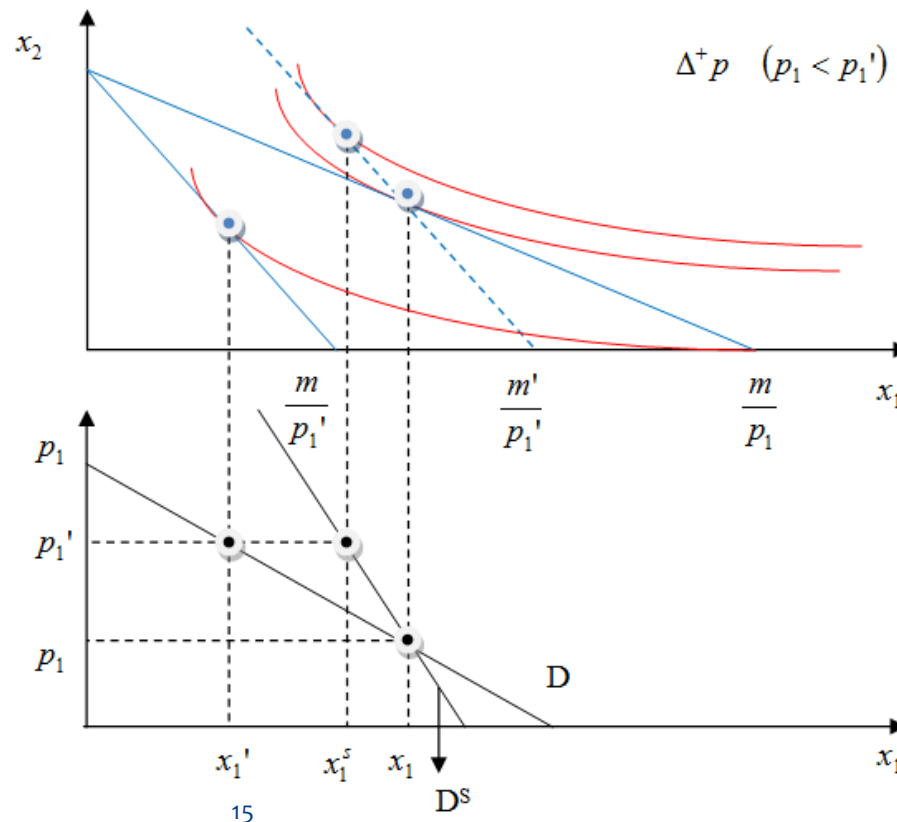
Demand functions

- * Standard demand function:
- * Income is kept constant.



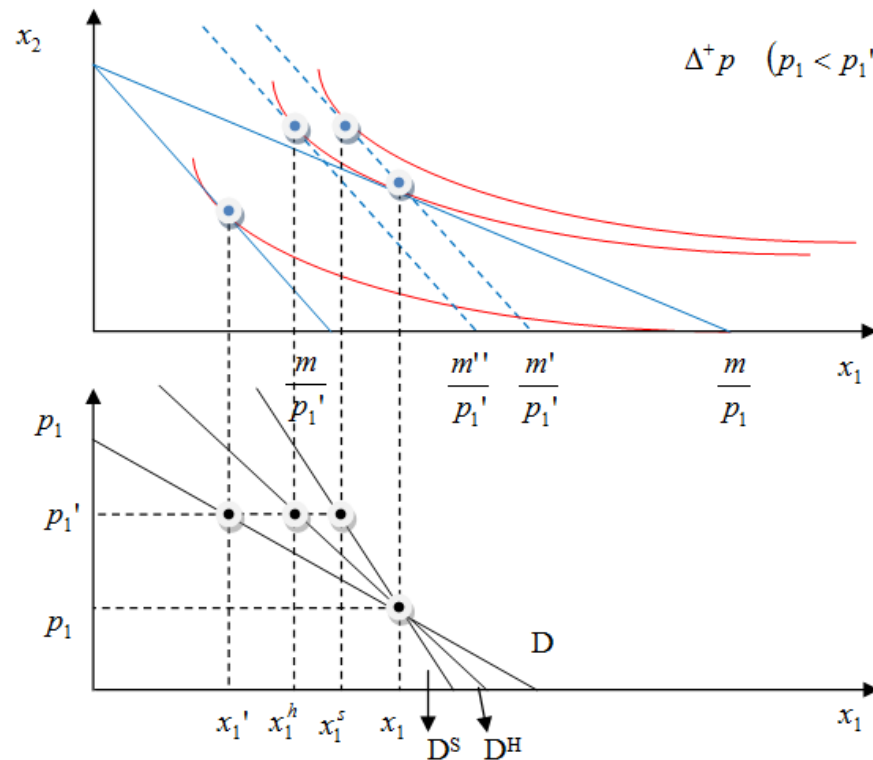
Demand functions

- * Slutsky demand function:
- * Purchasing power is kept constant.



Demand functions

- * Hicks demand function (or compensated demand curve):
- * Utility is kept constant.



Slutsky identity with derivatives

- * Standard demand function

$$x_1(p_1, p_2, m)$$

- * Slutsky demand function (with compensated income)

$$x_1^s(p_1, p_2, \bar{x}_1, \bar{x}_2)$$

- * Keep $(\bar{x}_1, \bar{x}_2)$ constant and compensate income so that this bundle is always on the budget constraint. In this point, the demand functions are identical. $x_1^s(p_1, p_2, \bar{x}_1, \bar{x}_2) \equiv x_1(p_1, p_2, \bar{m}(p_1, p_2))$

$$\bar{m} = p_1 \bar{x}_1 + p_2 \bar{x}_2$$

- * Apply the chain rule of differentiation:

$$\frac{\partial x_1^s(p_1, p_2, \bar{x}_1, \bar{x}_2)}{\partial p_1} \equiv \frac{\partial x_1(p_1, p_2, \bar{m})}{\partial p_1} + \frac{\partial x_1(p_1, p_2, \bar{m})}{\partial m} \frac{\partial \bar{m}(p_1, p_2)}{\partial p_1}$$



Slutsky identity with derivatives

$$\frac{\partial x_1^s(p_1, p_2, \bar{x}_1, \bar{x}_2)}{\partial p_1} \equiv \frac{\partial x_1(p_1, p_2, \bar{m})}{\partial p_1} + \frac{\partial x_1(p_1, p_2, \bar{m})}{\partial m} \frac{\partial \bar{m}(p_1, p_2)}{\partial p_1} \Leftrightarrow$$
$$\Leftrightarrow \frac{\partial x_1(p_1, p_2, \bar{m})}{\partial p_1} \equiv \frac{\partial x_1^s(p_1, p_2, \bar{x}_1, \bar{x}_2)}{\partial p_1} - \frac{\partial x_1(p_1, p_2, \bar{m})}{\partial m} \bar{x}_1$$

- * Total effect: $\frac{\partial x_1(p_1, p_2, \bar{m})}{\partial p_1}$
- * Substitution effect: $\frac{\partial x_1^s(p_1, p_2, \bar{x}_1, \bar{x}_2)}{\partial p_1}$
- * Income effect: $-\frac{\partial x_1(p_1, p_2, \bar{m})}{\partial m} \bar{x}_1$

